

Analytical Workflow for Malaria Disease Dynamics in India

Data Integration, Feature Engineering, Forecasting Models and Public-Health Interpretation

1. Objective and Scope

Malaria remains one of India's most geographically uneven infectious disease burdens, concentrated in forested, tribal and hard-to-reach districts. The task brief asked for an end-to-end workflow — data integration, spatio-temporal analysis and a forecasting model — applied to one disease. I chose malaria because India publishes multi-year, district-level case data and has an explicit national elimination target (2027 for zero indigenous transmission, 2030 for elimination), which gives the analysis a direct policy benchmark to evaluate against.

2. Data Sources

- **Case data** — district-wise malaria cases/deaths (2000–2024) and state-wise cases, blood slides tested, Pf cases and deaths (2022–2026), the latter extracted from a scanned NVBDCP/NCVBDC PDF annual report.
- **Environmental covariates** — elevation (SRTM DEM), surface-water occurrence (JRC Global Surface Water) and vegetation index (MODIS NDVI), pulled per district centroid via Google Earth Engine.
- **Socio-economic indicators** — state-level literacy, IMR, population density, per-capita NSDP, sanitation, safe-water access, banking access and rural tele-density ranks.
- **Geolocation** — district centroids geocoded (geopy/Nominatim) to align all rasters and tables on a common state–district key.

3. Pre-processing and Data Cleaning — What I Did and Why

Raw public-health data in India is rarely analysis-ready, so most of the effort went into cleaning before any modelling. Each step below was a deliberate decision, not a default pandas operation:

3.1 Extracting the state-wise data from a PDF report

- **What:** used pdfplumber to detect and extract every table on every page of the NVBDCP annual report PDF, generating unique column names for header cells that were blank or duplicated, then concatenated all extracted tables into one CSV.
- **Why:** the official annual report is only published as a formatted PDF, not a machine-readable file — this was the only way to get state-wise slide-positivity and Pf-case data into a usable table without manual re-typing.

3.2 Standardising text fields (state, district)

- **What:** stripped whitespace and applied consistent title-casing to every state and district string across all five source files before any merge.
- **Why:** the district-wise file, the state-wise file, the environmental extract and the socio-economic file each came from a different publisher, so the same place appeared as "WEST BENGAL", " West Bengal", and "west bengal " across files — a merge on raw strings would silently drop matching rows instead of raising an error.

3.3 Numeric type safety

- **What:** converted total_cases, total_deaths, blood_slides_tested, positive_cases, pf_cases and deaths to numeric using pd.to_numeric / string-clean-then-cast, stripping thousands-separator commas and stray spaces first.
- **Why:** the source tables store large case counts as text with commas (e.g. "12,340"), which pandas reads as an object/string column by default — left uncleaned, every downstream arithmetic feature (rates, lags, rolling means) would silently fail or produce NaN.

3.4 Handling the partial year ("2026 (upto June)")

- **What:** flagged such rows with an is_partial_year boolean instead of dropping them, and extracted a clean numeric year_clean via regex for merging/sorting.
- **Why:** a partial year is still informative for recent-trend context, but is not comparable to a full year's total — keeping it as a flagged column instead of silently truncating or discarding it lets the model exclude it exactly where it would bias training (e.g. lag/rolling features) while still using it for reporting.

3.5 Recovering "Unknown" states in the district file

- **What:** built a district→state lookup dictionary from the geocoded district list and used it to fill rows where the state was recorded as "Unknown", keeping a before/after flag (state_is_unknown_original vs state_is_unknown) and exporting the rows that remained unresolved for manual audit rather than deleting them.

- **Why:** silently dropping "Unknown" rows would have thrown away real case counts and biased state-level totals downward; recovering the state from the district name preserves the data while making the imputation fully auditable.

3.6 Reconciling district name fragmentation

- **What:** maintained an explicit rename dictionary for known variants (e.g. East Nimar → Khandwa, Ahmadabad → Ahmedabad, Osmanabad → Dharashiv reflecting the 2022 renaming, Nellore → Spsr Nellore), then re-aggregated (summed) any district-year rows that collided after standardisation, since these usually reflect the same district reported under two names in different years.
- **Why:** Indian districts are frequently renamed, split or merged (bifurcation is common); without this step the same physical district would appear as two separate, shorter time series, breaking every lag/rolling/trend feature computed per district.

3.7 Sanity-check flags instead of silent deletion

- **What:** added boolean flags for internally inconsistent rows — positive cases exceeding blood slides tested, Pf cases exceeding total positive cases, and deaths exceeding total cases — and reported counts rather than auto-correcting or dropping them.
- **Why:** these are classic manual-entry or table-extraction errors; auto-fixing them risks masking real data-quality issues, so they are surfaced for inspection and the analyst (or programme) can decide the right correction.

3.8 Duplicate handling

- **What:** dropped exact duplicate state-year rows; for district-year duplicates created by the renaming step, aggregated by summing rather than dropping.
- **Why:** duplicates from re-export are noise and should be removed, but duplicates created by name harmonisation represent real, different case counts that need to be combined, not discarded — treating both the same way would have either double-counted cases or lost them.

3.9 Merging static environmental data with a time series

- **What:** merged the single-year (2024) elevation / water-occurrence / NDVI extract onto the district panel by district only (not by year), and explicitly logged the match rate (share of rows with non-null NDVI) after the merge.
- **Why:** elevation and long-run water occurrence are effectively static properties of a district, so a year-key merge would be wrong; logging the match rate catches silent merge failures caused by any district name that still doesn't line up after step 3.6.

4. Feature Engineering — What Was Built and Why It Is Useful

Features were engineered separately for the district panel (annual, 2000–2024) and the state panel (2022–2026), each designed to give the model information that would genuinely have been available before the year being predicted.

4.1 District-level features

- **cases_lag1 / lag2 / lag3** — the district's own case count in the previous 1–3 years. Malaria transmission is locally self-sustaining (an established vector population and parasite reservoir), so recent history is normally the single strongest predictor of near-term burden.
- **cases_rol13 / cases_rol5** — 3- and 5-year rolling means (computed on prior years only, via shift-then-rolling) that smooth out single-year reporting noise or a one-off outbreak, giving the model a more stable baseline than the last value alone.
- **cases_trend_slope** — the slope of a linear fit through all prior years' cases at each point in time, capturing whether a district is on a genuine multi-year decline or incline — this is the feature that most directly encodes "is intervention working here."
- **cases_historical_mean** — the expanding (all-prior-years) mean, i.e. whether a district is historically a high- or low-burden area regardless of the current year's fluctuation.
- **cases_volatility_5yr** — rolling 5-year standard deviation of cases, flagging districts with erratic, spike-prone reporting versus stable trends — useful both as a model feature and as a data-quality signal.
- **elevation_m, water_occurrence_pct, ndvi_2024** — the ecological drivers of vector breeding habitat and survival (see Section 8), included as they change slowly and provide a baseline suitability signal independent of the case history.
- **high_water_area, high_ndvi** — binarised versions of the above (above-median flags), added because tree-based models can sometimes exploit a clean threshold split more efficiently than a continuous ramp, and because these thresholds are directly interpretable ("is this a high-standing-water district, yes/no").
- **Excluded for leakage** — `rank_within_state_year` (a same-year rank of the target) and `cases_per_1000m_elevation` (a same-year ratio using the target) were deliberately left out of the model's feature list even though they exist in the cleaned file, because both are computed from the current year's `total_cases` — including them would let the model see a transformed version of the answer it is predicting.

4.2 State-level features

- **positive_cases_lag1, positive_cases_rol2** — the state-level equivalent of the district lag/rolling features (a 2-year window, since only a handful of full years exist at state level).
- **testing_intensity** — blood slides tested that year relative to the state's own historical average testing volume — added so the model can partly distinguish a genuine case increase from a testing-effort increase (more testing mechanically finds more cases).
- **Socio-economic rank features** — per-capita NSDP, population density, literacy, IMR, rural tele-density, banking access, latrine facility and safe-water ranks, included because they proxy a state's health-system capacity and living conditions, both of which affect exposure and case detection.
- **Excluded for leakage** — `slide_positivity_rate`, `pf_percentage`, `case_fatality_rate` and `cases_yoy_change_pct` were all computed directly from `positive_cases` in the same row and were excluded from the state model's feature list for the same reason as above.

Two derived indicators — slide positivity rate (SPR) and case-fatality rate (CFR) — were deliberately kept in the cleaned dataset for reporting and EDA even though they are excluded from the model inputs: SPR is the standard WHO/NVBDCP severity indicator (positive slides per 100 tested) and CFR tracks case-management quality, so both are useful to report on even though neither may be used to predict the very count they are derived from.

5. Exploratory Analysis — What the Charts Show

The EDA layer runs across the merged district and state panels; the eight charts referenced below are the ones actually produced by the pipeline.

- **Top 20 districts by mean annual cases (2000–2024):** the ranking is dominated by Odisha (Rayagada, Kalahandi, Koraput, Kandhamal, Malkangiri, Kendujhar, Nabarangpur, Mayurbhanj, Ganjam, Sambalpur, Anugul), the Chhattisgarh–Jharkhand belt (Bastar, Palamu, West Singhbhum, Jashpur, Surguja, Latehar) and Kolkata, which tops the list. This matches the well-documented concentration of India's malaria burden in the eastern/central tribal, forested corridor — exactly the belt targeted by IMEP-3.
- **National trend, 2000–2024:** cases hover near 1.3–1.5 million through the early-to-mid 2000s, plateau through the early 2010s, then decline sharply and fairly steadily from about 2017 onward to under 0.2 million by 2023–24 — visually

consistent with NVBDCP's reported ~97% fall in annual cases since independence and India's 2024 exit from the WHO High-Burden-to-High-Impact group.

- **Pre-2016 vs. 2016-onward boxplot:** the pre-2016 distribution sits around 1.0–1.2 million with a fairly wide spread, while the 2016-onward distribution collapses to roughly 0.1–0.3 million with one high outlier — a visibly large level shift right at the point the National Framework for Malaria Elimination was launched, which is what the accompanying t-test is designed to confirm formally rather than just visually.
- **Improved (declining) vs. still-high-burden districts:** of the districts flagged, the "still high burden" group dominates the chart — West Singhbhum, Dhalai, Lawngtlai, Lunglei, Godda, Mamit, Pakur, Murshidabad, Hardoi, Howrah, and the 24 Parganas districts all show high 2022–2024 mean case counts with little recent improvement, while very few districts (Upper Subansiri, Karbi Anglong, and similar) appear in the "improved" category at comparable case volumes. This is arguably the most policy-relevant chart in the analysis: it shows the national decline is not uniform — a fairly small set of districts, concentrated in the Northeast (Tripura, Mizoram), eastern West Bengal, and the Jharkhand belt, still carry a disproportionate and largely un-declining share of the remaining burden, which is precisely the profile of the 159 districts IMEP-3 is designed to target.
- **Environmental profile, low vs. high burden districts:** elevation is lower and less variable in high-burden districts than in low-burden ones (malaria transmission requires warm low-to-mid elevation); water occurrence rises from low- to high-burden groups (more standing/surface water, more breeding habitat); NDVI is broadly similar across groups with a mild decline from low to high burden — a weaker signal than expected, most likely because a single 2024 vegetation snapshot cannot capture 24 years of case history well.
- **Case study — Bhilwara vs. Narayanpet:** these two districts were matched because they have near-identical environmental profiles (elevation, water occurrence, NDVI) but very different case trajectories — Bhilwara peaked near 8,000 cases in the early 2000s before declining steeply to near zero, while Narayanpet stayed low throughout. Since the environment is held roughly constant, the gap is attributable to non-environmental factors — historical intervention intensity, healthcare access, or reporting practices — which is exactly the kind of finding a purely environmental model would miss.
- **Correlation — district cases vs. environmental drivers:** at the aggregate level the correlations are weak (elevation –0.04, water occurrence 0.12, NDVI –0.02 against total_cases). This is expected, not a failure: a single static 2024 snapshot is being correlated against 24 years of case history, and the true within-district relationship (captured properly by the model's lag/trend features) is diluted when everything is pooled into one cross-sectional correlation.
- **Correlation — state cases vs. socio-economic ranks:** rural tele-density rank (0.42), per-capita NSDP rank (0.31) and IMR rank (0.29) show the strongest positive correlations with positive_cases, while latrine-facility rank (–0.43) shows the strongest negative correlation. Because these are rank variables (a lower rank number generally means a better outcome for that indicator), the signs need to be read carefully — the practical reading is that states with poorer sanitation infrastructure and higher infant mortality tend to also carry a higher malaria burden, consistent with malaria being concentrated in the same structurally under-resourced states rather than being an isolated public-health issue.

6. Forecasting Models and Results

Three models were built and evaluated against each other and against a naive last-year-value baseline, using MAE, RMSE and MAPE.

Model	Scale / Group	MAE	RMSE	MAPE %
District — GradientBoosting	District, 2024 test	183.8	941.2	983.4
District — Naive (last year)	District, 2024 test	260.5	1,294.7	1,429.9
State — GBM (LOYO CV mean)	State, CV mean	2,265.1	5,273.3	—
Combined — Stacked/weighted blend	State-year overlap	2,201.9	4,343.2	640.5
Top-down — State model (direct)	State-year overlap	3,214.8	7,960.4	169.7
Combined — Simple average	State-year overlap	3,237.0	7,717.3	101.4
Bottom-up — District model (aggregated)	State-year overlap	3,386.4	7,599.8	50.4

6.1 District model — beats the naive baseline

The Gradient Boosting model (log1p-transformed target, trained through 2023 and tested on 2024) reduced MAE by 29.4% and RMSE by 27.3% relative to simply predicting last year's case count. This is the cleanest evidence in the whole analysis that the

engineered lag/rolling/trend/environmental features carry real predictive signal beyond persistence — the model is not just "predicting last year again with extra steps."

6.2 Why MAPE is so large

Both district-level MAPE figures (983% for the model, 1,430% for naive) look alarming next to the MAE/RMSE columns, but this is a known artefact of MAPE on near-zero targets rather than a sign the model is unusable: by 2024, a large number of districts have genuinely declined to single-digit or zero case counts (visible directly in the national trend chart), and MAPE divides by the actual value — a district going from 2 cases to 8 is only a 6-case error (small MAE contribution) but a 300% error. MAE and RMSE, which are reported in absolute cases, are the more trustworthy metrics here and both clearly favour the model over the naive baseline.

6.3 State model — small-sample caution

The state-level model was evaluated with leave-one-year-out cross-validation because only four full years of state-wise data are available. A mean CV MAE of ~2,265 cases is reported honestly as indicative rather than robust — with only 3–4 states-years per fold, a single unusual state-year can move the fold's error substantially, so this number should not be quoted with the same confidence as the district-level test-set result.

6.4 Hierarchical reconciliation — the combined model wins

Aggregating the district model's predictions up to state totals ("bottom-up") and comparing them against the state model's own direct prediction ("top-down") for the same state-years shows the two disagree substantially — bottom-up alone actually performs worst on MAE/RMSE among the combined-model comparisons (3,386 / 7,600), while top-down and a simple average sit in between. The learned stacked blend of the two, however, achieves the best MAE and RMSE of all the state-scale approaches — 2,201.9 MAE and 4,343.2 RMSE — a 2.8% MAE and 17.6% RMSE improvement over the state model used alone. This is the key modelling result: forecast reconciliation adds value here, meaning the district-level and state-level signals are complementary rather than redundant, even though the district file's total_cases and the state file's positive_cases come from different reports and are not perfectly comparable in scale.

One caveat worth flagging on the same result: bottom-up has the lowest MAPE (50.4%) of the three individual/blend approaches despite the highest MAE/RMSE, meaning its errors are more consistent in relative terms even though larger in absolute terms — a useful sanity check that a single metric should not be read in isolation.

6.5 Supplementary Analysis — District Classification, Environmental Profile, Case-Study Pairs and State-Model Diagnostics

The underlying output tables behind Sections 5 and 6 (district improved/needs-action lists, the environmental burden profile, matched-pair case studies, state summary statistics, feature importance, and per-state-year residuals) support the qualitative findings above with the following specific figures.

- 15 districts are classified as **"improved"**, with pct-change from early-period to recent-period mean cases at or near **-100%** (e.g., Karbi Anglong: 13,504 to 0; East Siang: 10,966 to 0), while 15 districts are classified **"needs action"** with sharp increases: Mizoram-Lawngtlai **+4,118%**, Tripura-Dhalai **+690%**, West Bengal-Murshidabad **+1,429%**, West Bengal-Howrah **+1,424%**.
- The "needs action" list is geographically concentrated in **Tripura, Mizoram, Jharkhand and West Bengal** — the same eastern/Northeast belt already identified in Section 5 and targeted by IMEP-3, giving a district-level quantitative confirmation of that classification.
- The ideal-condition environmental profile shows a monotonic gradient across burden groups: elevation falls from **623.8m (low burden) to 332.3m (high burden)**, while water occurrence rises from **31.8% to 36.1%** — the strongest quantitative confirmation of the vector-biology argument in Section 8 (breeding requires standing water and warm, low elevation).
- Matched environmental case-study pairs extend the Bhilwara-Narayanpet example already discussed: Odisha-Anugul vs. its environmental twin Tirunelveli (TN) shows a **176x gap** in mean cases despite near-identical terrain; Gujarat-Kheda vs. Bihar-Madhubani shows a **174x gap**. The strongest version of this argument is UP-Bhadohi vs. UP-Azamgarh — two districts in the **same state** with near-identical environment but a **145x gap** in case counts, which rules out state-level policy or reporting differences as the explanation.
- State-level summary statistics show extreme concentration of burden: **Odisha, Jharkhand and Chhattisgarh alone account for roughly 9.26 million of the cumulative case total** in the state panel, with Odisha the single largest contributor (~4.34M, mean 6,893 cases/year).
- State-model feature importance confirms the model is driven almost entirely by case momentum: **positive_cases_lag1 (62.0%) and positive_cases_roll2 (36.9%) jointly account for 98.9%** of importance, while testing_intensity contributes only **1.07%** — socio-economic and testing-effort signal adds negligible marginal value at state level, a limitation worth naming explicitly alongside Section 9.

- Per-state-year residual analysis of the CV predictions shows the model improved between folds: mean absolute error was **~3,053 cases in the 2024 fold vs. ~1,477 in the 2025 fold**. The largest single errors in the entire table are both in Odisha — **+36,908 (2024) and +11,455 (2025)**, both over-predictions — followed by Jharkhand 2024 (+22,021) and Maharashtra 2024 (+11,001). The largest under-predictions (missed surges) were Uttar Pradesh 2025 (-6,531), Tripura 2025 (-5,603) and Mizoram 2025 (-5,142).
- Because Odisha is simultaneously the **highest-burden state and the source of the model's largest absolute errors in both years**, this is the single most policy-relevant diagnostic in the supplementary tables: forecast uncertainty is largest exactly where the case burden — and therefore the stakes of an inaccurate forecast — is highest.
- Best-predicted states (smallest residuals) were consistently very-low-case states — Lakshadweep, Arunachal Pradesh, Punjab, Sikkim — which is consistent with the MAPE-instability caveat already raised in Section 6.2 for near-zero targets.

7. Government Action and How the Analysis Connects to It

India's control effort is centred on the National Framework for Malaria Elimination (NFME, 2016–2030), which stratifies states into intensified-control, pre-elimination, elimination and prevention-of-re-establishment categories, and the National Strategic Plan for Malaria Elimination 2023–2027, which layers on real-time surveillance through the Integrated Health Information Platform, a 'test-treat-track' case protocol, Integrated Vector Management, and targeted LLIN (bed-net) distribution. The Intensified Malaria Elimination Project (IMEP-3) currently focuses on 159 high-burden districts across 12 states. Reported annual cases have fallen from roughly 7.5 crore at independence to about 20 lakh by 2023, and India exited the WHO's High Burden to High Impact group in 2024.

My analysis is a natural companion to this programme structure: the pre/post-2016 comparison directly tests whether the NFME's launch coincides with a measurable shift in the national trend; the district trend-classification ('improved' vs. 'still high burden') mirrors the NFME's own category system and lines up closely with the districts IMEP-3 already targets; and the forecasting layer — particularly the reconciled district+state combined forecast — offers an early-warning signal that complements IHIP's real-time reporting rather than replacing it, which matters most in the remote, forested districts that both this analysis and the national programme identify as still carrying the bulk of the burden.

8. Link to Malaria's Lifecycle — Why These Features Were Chosen

Malaria transmission is a two-host cycle: Plasmodium parasites develop inside an Anopheles mosquito (sporogonic cycle, temperature- and humidity-dependent, roughly 10–18 days) before being transmitted to a human via bite, followed by the parasite's liver and blood stages in the human host. This biology is why the environmental features carry predictive signal: standing/slow-moving water (JRC water-occurrence) is the mosquito's larval breeding habitat; vegetation cover and humidity (NDVI) sustain adult mosquito survival long enough to complete the sporogonic cycle; and elevation constrains temperature, gating whether the parasite can complete development in the vector at all — this is why high-altitude districts are structurally low-risk regardless of other conditions, and it is exactly the pattern the low-vs-high-burden elevation boxplot shows. The lag and rolling-average case features implicitly capture the human–mosquito–human transmission cycle's persistence: a district with sustained transmission in prior years has an established local vector population and parasite reservoir, making it more likely to sustain transmission going forward, whereas a sharp environmental or programmatic change (post-monsoon water recession, an LLIN campaign) can break that cycle — which is exactly what the trend-slope feature and the improved-vs-needs-action classification are built to detect.

9. Limitations and Assumptions

- Case counts are passively reported (health-facility detected), not true incidence; the model predicts reported burden, not underlying transmission.
- Environmental covariates are single-year/static snapshots (2024) applied across a 24-year case series — they describe typical district ecology, not year-specific conditions, which likely explains the weak raw environmental correlations in Section 5.
- The state-level model has only 4 full years of data, so its cross-validated metrics are indicative rather than robust.
- MAPE is unstable and should be read alongside MAE/RMSE wherever recent case counts approach zero, as in the district test set.
- Google Mobility and finer-grained socio-economic data were considered but not fully integrated in this version — a natural next step for capturing human-movement-driven spread.

10. Conclusion

The pipeline demonstrates a reproducible path from raw, heterogeneous public-health data — including a scanned PDF report — to a leakage-checked, two-scale forecasting model with an explicit, honestly-reported evaluation against naive baselines. The district-level model beats the naive baseline by ~29% on MAE, and the hierarchical reconciliation between district and state forecasts further improves on either model alone, showing the two data sources are complementary. Framing these results around India's actual elimination milestones and the underlying vector biology grounds the analysis in a real decision context — identifying where the national programme's interventions appear to be working, where burden is not budging, and where a forecast could give NVBDCP/NCVBDC useful lead time ahead of the 2027 zero-indigenous-transmission target.